

Mark Scheme

Q1.

Part	Working or answer an examiner might expect to see	Mark	Notes
	$\lim_{\delta x \rightarrow 0} \sum_{x=4}^9 \sqrt{x} \delta x = \int_4^9 \sqrt{x} \, dx$	B1	This mark is given for writing the expression for a sum as an integral
	$\left[\frac{2}{3} x^{\frac{3}{2}} \right]_4^9 = \frac{2}{3} \times 9^{\frac{3}{2}} - \frac{2}{3} \times 4^{\frac{3}{2}}$	M1	This mark is given for a method to evaluate the integral
	$= \frac{38}{3}$	A1	This mark is given for a correct evaluation of the integral
(Total 3 marks)			

Q2.

Question	Scheme	Marks	AOs
(a)	$\lim_{\delta x \rightarrow 0} \sum_{x=2.1}^{6.3} \frac{2}{x} \delta x = \int_{2.1}^{6.3} \frac{2}{x} \, dx$	B1	1.2
		(1)	
(b)	$= [2 \ln x]_{2.1}^{6.3} = 2 \ln 6.3 - 2 \ln 2.1$	M1	1.1b
	$= \ln 9 \quad \text{CSO}$	A1	1.1b
		(2)	
(3 marks)			
Notes:			

Mark (a) and (b) as one

(a)

B1: States that $\int_{2.1}^{6.3} \frac{2}{x} dx$ or equivalent such as $2 \int_{2.1}^{6.3} x^{-1} dx$ but must include the limits and the dx .

Condone $dx \leftrightarrow \delta x$ as it is very difficult to tell one from another sometimes

(b)

M1: Know that $\int \frac{1}{x} dx = \ln x$ and attempts to apply the limits (either way around)

Condone $\int \frac{2}{x} dx = p \ln x$ (including $p = 1$) or $\int \frac{2}{x} dx = p \ln qx$ as long as the limits are applied.

Also be aware that $\int \frac{2}{x} dx = \ln x^2$, $\int \frac{2}{x} dx = 2 \ln|x| + c$ and $\int \frac{2}{x} dx = 2 \ln cx$ o.e. are also correct

$[p \ln x]_{2.1}^{6.3} = p \ln 6.3 - p \ln 2.1$ is sufficient evidence to award this mark

A1: CSO $\ln 9$. Also answer = $\ln 3^2$ so $k = 9$ is fine. Condone $\ln|9|$

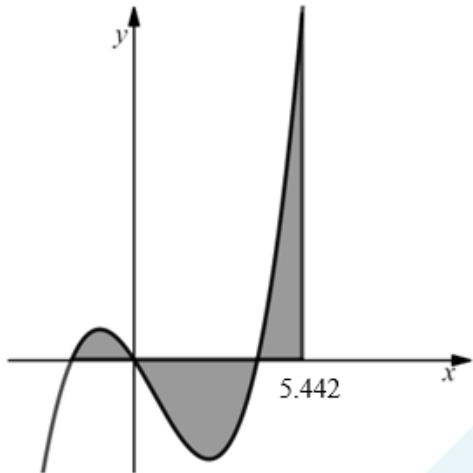
The method mark must have been awarded. Do not accept answers such as $\ln \frac{39.69}{4.41}$

Note that solutions appearing from "rounded" decimal work when taking lns should not score the final mark. It is a "show that" question

E.g. $[2 \ln x]_{2.1}^{6.3} = 2 \ln 6.3 - 2 \ln 2.1 = 2.197 = \ln k \Rightarrow k = e^{2.197} = 8.998 = 9$

Q3.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$y = x(x+2)(x-4) = x^3 - 2x^2 - 8x$	B1	This mark is given for expanding brackets as a first step to a solution
	$\int_{-2}^0 x^3 - 2x^2 - 8x \, dx$	M1	This mark is given for a method to find the exact area of R_1
	$= \left[\frac{1}{4}x^4 - \frac{2}{3}x^3 - 4x^2 \right]_{-2}^0$	M1	This mark is given for a method to evaluate the integral
	$= 0 - (4 - \frac{-16}{3} - 16) = \frac{20}{3}$	A1	This mark is given for a full method to show the exact value of R_1
(b)	$\frac{1}{4}b^4 - \frac{2}{3}b^3 - 4b^2 = -\frac{20}{3}$	M1	This mark is given for deducing the area of $R_2 = -\frac{20}{3}$
	$3b^4 - 8b^3 - 48b^2 + 80 = 0$	A1	This mark is given for rearranging the equation to a quartic
	$(b+2)^2(3b^2 - 20b + 20)$ $= (b^2 + 4b + 4)(3b^2 - 20b + 20)$	M1	This mark is given for expanding the equation given
	$= 3b^4 - 8b^3 - 48b^2 + 80 = 0$ The two equations are the same, so verified	A1	This mark is for showing, and stating, that the equations are the same

(c)		B1	This mark is given for a sketch of the curve with $b = 5.442$ shown
	Between $x = -2$ and $b = 5.442$, the area above the x -axis is the same as the area below the x -axis	B1	This mark is given for a valid explanation of the significance of the root 5.442

Q4.

Question	Scheme	Marks	AOs
(a)	$\frac{dr}{dt} \propto \pm \frac{1}{r^2}$ or $\frac{dr}{dt} = \pm \frac{k}{r^2}$ (for k or a numerical k)	M1	3.3
	$\int r^2 dr = \int \pm k dt \Rightarrow \dots$ (for k or a numerical k)	M1	2.1
	$\frac{1}{3}r^3 = \pm kt \{+ c\}$	A1	1.1b
	$t = 0, r = 5$ and $t = 4, r = 3$ gives $\frac{1}{3}r^3 = -\frac{49}{6}t + \frac{125}{3}$, where r , in mm, is the radius {of the mint} and t , in minutes, is the time from when it {the mint} was placed in the mouth	M1	3.1a
	$t = 0, r = 5$ and $t = 240, r = 3$ gives $\frac{1}{3}r^3 = -\frac{49}{360}t + \frac{125}{3}$, where r , in mm, is the radius {of the mint} and t , in seconds, is the time from when it {the mint} was placed in the mouth	A1	1.1b
		(5)	
(b)	$r = 0 \Rightarrow 0 = -\frac{49}{6}t + \frac{125}{3} \Rightarrow 0 = -49t + 250 \Rightarrow t = \dots$	M1	3.4
	time = 5 minutes 6 seconds	A1	1.1b
		(2)	
(c)	Suggests a suitable limitation of the model. E.g. - Model does not consider how the mint is sucked - Model does not consider whether the mint is bitten - Model is limited for times up to 5 minutes 6 seconds, o.e. - Not valid for times greater than 5 minutes 6 seconds, o.e. - Mint may not retain the shape of a sphere (or have uniform radius) as it is being sucked - The model indicates that the radius of the mint is negative after it dissolves - Model does not consider the temperature in the mouth - Model does not consider rate of saliva production - Mint could be swallowed before it dissolves in the mouth	B1	3.5b
		(1)	
(8 marks)			

Notes for Question	
(a)	
M1:	Translates the description of the model into mathematics. See scheme.
M1:	Separates the variables of their differential equation which is in the form $\frac{dr}{dt} = f(r)$ and some attempt at integration. (e.g. attempts to integrate at least one side). e.g. $\int r^2 dr = \int \pm k dt$ and some attempt at integration. Condone the lack of integral signs
Note:	You can imply the M1 mark for $r^2 dr = -k dt \Rightarrow \frac{1}{3}r^3 = -kt$
Note:	A numerical value of k (e.g. $k = \pm 1$) is allowed for the first two M marks
A1:	Correct integration to give $\frac{1}{3}r^3 = \pm kt$ with or without a constant of integration, c
M1:	For a complete process of using the boundary conditions to find both their unknown constants and finds an equation linking r and t So applies either $t = 0, r = 5$ and $t = 4, r = 3$, or $t = 0, r = 5$ and $t = 240, r = 3$, on their integrated equation to find their constants k and c and obtains an equation linking r and t
A1:	Correct equation, with variables r and t fully defined including correct reference to units. $\frac{1}{3}r^3 = -\frac{49}{6}t + \frac{125}{3}$, {or an equivalent equation,} where r, in mm, is the radius {of the mint} and t, in minutes, is the time from when it {the mint} was placed in the mouth $\frac{1}{3}r^3 = -\frac{49}{360}t + \frac{125}{3}$, {or an equivalent equation,} where r, in mm, is the radius {of the mint} and t, in seconds, is the time from when it {the mint} was placed in the mouth
Note:	Allow correct equations such as - in minutes, $r = \sqrt[3]{\frac{250-49t}{2}}$, $r^3 = -\frac{49}{2}t + 125$ or $t = \frac{250-2r^3}{49}$ - in seconds, $r = \sqrt[3]{\frac{15000-49t}{120}}$, $r^3 = -\frac{49}{120}t + 125$ or $t = \frac{15000-120r^3}{49}$
Note:	t defined as "the time from the start" is not sufficient for the final A1

(b)	
M1:	Sets $r = 0$ in their part (a) equation which links r with t and rearranges to make $t = \dots$
A1:	5 minutes 6 seconds cao (Note: 306 seconds with no reference to 5 minutes 6 seconds is A0)
Note:	Give M0 if their equation would solve to give a negative time or a negative time is found
Note:	You can mark part (a) and part (b) together
(c)	
B1:	See scheme
Note:	Do not accept by itself - mint may not dissolve at a constant rate - rate of decrease of mint must be constant $0 \leq t < \frac{250}{49}$, $r \geq 0$; without any written explanation - reference to a mint having $r > 5$

Q5.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	The asymptote is found where $2x - q = 0$ Hence $q = 4$	B1	This mark is given for explaining that the asymptote at $x = 2$ is a solution of $2x - q = 0$
	$y = \frac{p-3x}{(2x-4)(x+3)}$ $\frac{1}{2} = \frac{p-9}{(6-4)(3+3)}$	M1	This mark is given for substituting $x = 3$, $y = \frac{1}{2}$ (and $q = 4$)

(b)	$\frac{15-3x}{(2x-4)(x+3)} = \frac{A}{(2x-4)} + \frac{B}{(x+3)}$	M1	This mark is given for a method to use partial fractions
	$= \frac{1.8}{(2x-4)} - \frac{2.4}{(x+3)}$	M1	This mark is given for finding values for A and B
	$= \frac{0.9}{(x-2)} - \frac{2.4}{(x+3)}$	A1	This mark is given for a fully simplified expression
	$I = \int \frac{15-3x}{(2x-4)(x+3)} dx$ $= m \ln(2x-4) + n \ln(x+3)$	M1	This mark is given for a method to integrate to find the area of R
	$= 0.9 \ln(2x-4) + 2.4 \ln(x+3)$	A1	This mark is given for a correct expression for the area of R
	$\text{Area } R = \left[0.9 \ln(2x-4) - 2.4 \ln(x+3) \right]_3^5$	M1	This mark is given for deducing an expression for the area of R ($y = 0$ when $x = 5$)
	$= [0.9 \ln 6 - 2.4 \ln 8] - [0.9 \ln 2 - 2.4 \ln 6]$ $= [0.9 \ln 6 + 2.4 \ln 6] - [7.2 \ln 2 + 0.9 \ln 2]$ $= 3.3 \ln 6 - 8.1 \ln 2$ $= 3.3 \ln 3 + 3.3 \ln 2 - 8.1 \ln 2$	M1	This mark is given for a method to find the exact area of R
	$= 3.3 \ln 3 - 4.8 \ln 2$	A1	This mark is given for a correct value of the area of R with $a = 3.3$ and $b = 4.8$

Q6.

Question	Scheme	Marks	AOs
	Writes $\int \frac{t+1}{t} dt = \int 1 + \frac{1}{t} dt$ and attempts to integrate	M1	2.1
	$= t + \ln t (+c)$	M1	1.1b
	$(2a + \ln 2a) - (a + \ln a) = \ln 7$	M1	1.1b
	$a = \ln \frac{7}{2}$ with $k = \frac{7}{2}$	A1	1.1b
(4 marks)			
Notes:			
M1: Attempts to divide each term by t or alternatively multiply each term by t^{-1} M1: Integrates each term and knows $\int \frac{1}{t} dt = \ln t$. The $+c$ is not required for this mark M1: Substitutes in both limits, subtracts and sets equal to $\ln 7$ A1: Proceeds to $a = \ln \frac{7}{2}$ and states $k = \frac{7}{2}$ or exact equivalent such as 3.5			

Q7.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$dh = -2(4 - u) du$	B1	This mark is given for finding an expression for dh
	$\int \frac{dh}{4 - \sqrt{h}} = \int \frac{-2(4 - u) du}{4 - \sqrt{h}}$	M1	This mark is given for substituting $u = 4 - \sqrt{h}$ into the integral
	$= \int -\frac{8}{u} + 2 du$	M1	This mark is given for a method to find a simplified version of the integral
	$-8 \ln u + 2u + c$ $= -8 \ln (4 - \sqrt{h}) + 2(4 - \sqrt{h}) + c$	M1	This mark is given for integrating with respect to u to produce an expression in terms of h
		A1	This mark is given for a correct expression for the integral
	$= -8 \ln (4 - \sqrt{h}) - 2\sqrt{h} + k$	A1	This mark is given for a full proof to arrive at the answer as shown (appreciating that $k = c + 8$)
(b)	$\frac{dh}{dt} = 0 \Rightarrow 4 - \sqrt{h} = 0$	M1	This mark is given for a setting $\frac{dh}{dt} = 0$
	$0 < h < 16$	A1	This mark is given for deducing the range of the heights of the trees for this model

(c)	$\frac{dh}{dt} = \frac{t^{0.25}(4-\sqrt{h})}{20} \Rightarrow \frac{dh}{(4-\sqrt{h})} = \frac{t^{0.25}dt}{20}$	B1	This mark is given for separating the variables
	$-8 \ln(4-\sqrt{h}) - 2\sqrt{h} + k = \frac{t^{1.25}}{25}$	M1	This mark is given for a method to integrate both sides of the equation
		A1	This mark is given for integrating both sides of the equation correctly
	When $t = 0$ and $h = 1$, $-8 \ln 3 - 2 + k = 0$ $k = 2 + 8 \ln 3$	M1	This mark is given for substituting values of $t = 0$ and $h = 1$ to find a value for k
	When $h = 12$, $-8 \ln(4-\sqrt{12}) - 2\sqrt{12} + 2 + 8 \ln 3 = \frac{t^{1.25}}{25}$	M1	This mark is given for a method to find a value for t by substituting $h = 12$ into the equation
	$t^{1.25} = 221.2795 \Rightarrow t = \sqrt[1.25]{221.2795}$	M1	This mark is given for simplifying to find an expression for t
	$t = 75.2$ years	A1	This mark is given for correctly finding the time the tree would take to reach a height of 12 metres
(Total 15 marks)			

Q8.

Via firstly integrating

Question	Scheme	Marks	AOs
	$f'(x) = 6x^2 + ax - 23 \Rightarrow f(x) = 2x^3 + \frac{1}{2}ax^2 - 23x + c$	M1 A1	1.1b 1.1b
	"c" = -12	B1	2.2a
	$f(-4) = 0 \Rightarrow 2 \times (-4)^3 + \frac{1}{2}a(-4)^2 - 23(-4) - 12 = 0$	dM1	3.1a
	$a = \dots (6)$	dM1	1.1b
	$(f(x) =) 2x^3 + 3x^2 - 23x - 12$ Or Equivalent e.g. $(f(x) =) (x+4)(2x^2 - 5x - 3) \quad (f(x) =) (x+4)(2x+1)(x-3)$	A1cso	2.1
		(6)	
(6 marks)			

Notes:

M1: Integrates $f'(x)$ with two correct indices. There is no requirement for the $+c$

A1: Fully correct integration (may be unsimplified). The $+c$ must be seen (or implied by the -12)

B1: Deduces that the constant term is -12

dM1: Dependent upon having done some integration. It is for setting up a linear equation in a by using $f(-4) = 0$
 May also see long division attempted for this mark. Need to see a complete method leading to a remainder in terms of a which is then set $= 0$.

For reference, the quotient is $2x^2 + \left(\frac{a}{2} - 8\right)x + 9 - 2a$ and the remainder is $8a - 48$

May also use $(x+4)(px^2 + qx + r) = 2x^3 + \frac{1}{2}ax^2 - 23x - 12$ and compare coefficients to find p , q and r and hence a . Allow this mark if they solve for p , q and r

Note that some candidates use $2f(x)$ which is acceptable and gives the same result if executed correctly.

dM1: Solves the linear equation in a or uses p , q and r to find a .

It is dependent upon having attempted some integration and used $f(\pm 4) = 0$ or long division/comparing coefficients with $(x+4)$ as a factor.

A1cso: For $f(x) = 2x^3 + 3x^2 - 23x - 12$ oe. Note that " $f(x) =$ " does not need to be seen and ignore any " $= 0$ "

Via firstly using factor

Question	Scheme	Marks	AOs
Alt	$f(x) = (x+4)(Ax^2 + Bx + C)$	M1 A1	1.1b 1.1b
	$f(x) = Ax^3 + (4A+B)x^2 + (4B+C)x + 4C \Rightarrow C = -3$	B1	2.2a
	$f'(x) = 3Ax^2 + 2(4A+B)x + (4B+C)$ and $f'(x) = 6x^2 + ax - 23$ $\Rightarrow A = \dots$	dM1	3.1a
	Full method to get A , B and C	dM1	1.1b
	$f(x) = (x+4)(2x^2 - 5x - 3)$	A1cso	2.1
		(6)	
(6 marks)			

Notes:

M1: Uses the fact that $f(x)$ is a cubic expression with a factor of $(x+4)$

A1: For $f(x) = (x+4)(Ax^2 + Bx + C)$

B1: Deduces that $C = -3$

dM1: Attempts to differentiate either by product rule or via multiplication and compares to $f'(x) = 6x^2 + ax - 23$ to find A .

dM1: Full method to get A , B and C

A1cso: $f(x) = (x+4)(2x^2 - 5x - 3)$ or $f(x) = (x+4)(2x+1)(x-3)$

Q9.

Question	Scheme	Marks	AOs
(a)	$\frac{dH}{dt} = \frac{H \cos 0.25t}{40} \Rightarrow \int \frac{1}{H} dH = \int \frac{\cos 0.25t}{40} dt$	M1	3.1a
	$\ln H = \frac{1}{10} \sin 0.25t (+c)$	M1 A1	1.1b 1.1b
	Substitutes $t = 0, H = 5 \Rightarrow c = \ln(5)$	dM1	3.4
	$\ln\left(\frac{H}{5}\right) = \frac{1}{10} \sin 0.25t \Rightarrow H = 5e^{0.1 \sin 0.25t} *$	A1*	2.1
		(5)	
(b)	Max height = $5e^{0.1} = 5.53 \text{ m}$ (Condone lack of units)	B1	3.4
		(1)	
(c)	Sets $0.25t = \frac{5\pi}{2}$	M1	3.1b
	31.4	A1	1.1b
		(2)	
(8 marks)			

(a)

M1: Separates the variables to reach $\int \frac{1}{H} dH = \int \frac{\cos 0.25t}{40} dt$ or equivalent.

The integral signs need to be present on both sides and the dH AND dt need to be in the correct positions.

M1: Integrates both sides to reach $\ln H = A \sin 0.25t$ or equivalent with or without the $+c$

A1: $\ln H = \frac{1}{10} \sin 0.25t + c$ or equivalent with or without the $+c$. Allow two constants, one either side

If the 40 was on the lhs look for $40 \ln H = 4 \sin 0.25t + c$ or equivalent.

dM1: Substitutes $t = 0, H = 5 \Rightarrow c = \dots$ There needs to have been a single $+c$ to find.

It is dependent upon the previous M mark. You may allow even if you don't explicitly see $t = 0, H = 5$ as it may be implied from their previous line

If the candidate has attempted to change the subject and made an error/ slip then condone it for this M but not the final A. Eg. $40 \ln H = 4 \sin 0.25t + c \Rightarrow H^{40} = e^{4 \sin 0.25t} + e^c \Rightarrow 5^{40} = 1 + e^c \Rightarrow c = \dots$

Also many students will be attempting to get to the given answer so condone the method of finding $c = \dots$. These students will lose the A1* mark

A1*: Proceeds via $\ln H = \frac{1}{10} \sin 0.25t + \ln 5$ or equivalent to the given answer $H = 5e^{0.1 \sin 0.25t}$ with at least one correct intermediate line and no incorrect work.

DO NOT condone c 's going to c 's when they should be e^c or A

Accept as a minimum $\ln H = \frac{1}{10} \sin 0.25t + \ln 5 \Rightarrow H = e^{\frac{1}{10} \sin 0.25t + \ln 5}$ or $H = e^{\frac{1}{10} \sin 0.25t} \times e^{\ln 5}$ before sight of the given answer

If the only error was to omit the integration signs on line 1, thus losing the first M1, allow the candidate to have access to this mark following a correct intermediate line (see above).

If they attempt to change the subject first then the constant of integration must have been adapted if the A1* is to be awarded. $\ln H = \frac{1}{10} \sin 0.25t + c \Rightarrow H = e^{\frac{1}{10} \sin 0.25t + c} \Rightarrow H = Ae^{\frac{1}{10} \sin 0.25t}$

The dM1 and A1* under this method are awarded at virtually the same time.

Also, for the final two marks, you may see a proof from $\int_0^H \frac{40}{H} dH = \int_5^t \cos 0.25t dt$

.....
 There is an alternative via the use of an integrating factor.

(b)

B1: States that the maximum height is 5.53 m Accept $5e^{0.1}$ Condone a lack of units here, but penalise if incorrect units are used.

(c)

M1: For identifying that it would reach the maximum height for the 2nd time when $0.25t = \frac{5\pi}{2}$ or 450

A1: Accept awrt 31.4 or 10π Allow if units are seen

Q10.

Question	Scheme	Marks	AOs
	$\frac{dy}{dx} = \frac{15}{2}x^{\frac{1}{2}} - 9$	M1 A1	3.1a 1.1b
	Substitutes $x = 4 \Rightarrow \frac{dy}{dx} = 6$	M1	2.1
	Uses (4, 15) and gradient $\Rightarrow y - 15 = 6(x - 4)$	M1	2.1
	Equation of l is $y = 6x - 9$	A1	1.1b
	Area $R = \int_0^4 \left(5x^{\frac{3}{2}} - 9x + 11 \right) - (6x - 9) dx$	M1	3.1a
	$= \left[2x^{\frac{5}{2}} - \frac{15}{2}x^2 + 20x(+c) \right]_0^4$	A1	1.1b
	Uses both limits of 4 and 0 $\left[2x^{\frac{5}{2}} - \frac{15}{2}x^2 + 20x \right]_0^4 = 2 \times 4^{\frac{5}{2}} - \frac{15}{2} \times 4^2 + 20 \times 4 - 0$	M1	2.1
	Area of $R = 24$ *	A1*	1.1b
	Correct notation with good explanations	A1	2.5
		(10)	
(10 marks)			

Notes:

M1: Differentiates $5x^{\frac{3}{2}} - 9x + 11$ to a form $Ax^{\frac{1}{2}} + B$

A1: $\frac{dy}{dx} = \frac{15}{2}x^{\frac{1}{2}} - 9$ but may not be simplified

M1: Substitutes $x = 4$ in their $\frac{dy}{dx}$ to find the gradient of the tangent

M1: Uses their gradient and the point (4, 15) to find the equation of the tangent

A1: Equation of l is $y = 6x - 9$

M1: Uses Area $R = \int_0^4 \left(5x^{\frac{3}{2}} - 9x + 11 \right) - (6x - 9) dx$ following through on their $y = 6x - 9$

Look for a form $Ax^{\frac{5}{2}} + Bx^2 + Cx$

A1: $= \left[2x^{\frac{5}{2}} - \frac{15}{2}x^2 + 20x(+c) \right]_0^4$ This must be correct but may not be simplified

M1: Substitutes in both limits and subtracts

A1*: Correct area for $R = 24$

A1: Uses correct notation and produces a well explained and accurate solution. Look for

- Correct notation used consistently and accurately for both differentiation and integration
- Correct explanations in producing the equation of l . See scheme.
- Correct explanation in finding the area of R . In way 2 a diagram may be used.

Alternative method for the area using area under curve and triangles. (Way 2)

M1: Area under curve $= \int_0^4 \left(5x^{\frac{3}{2}} - 9x + 11 \right) = \left[Ax^{\frac{5}{2}} + Bx^2 + Cx \right]_0^4$

A1: $= \left[2x^{\frac{5}{2}} - \frac{9}{2}x^2 + 11x \right]_0^4 = 36$

M1: This requires a full method with all triangles found using a correct method

Look for Area $R = \text{their } 36 - \frac{1}{2} \times 15 \times \left(4 - \text{their } \frac{3}{2} \right) + \frac{1}{2} \times \text{their } 9 \times \text{their } \frac{3}{2}$

Q11.

Question	Scheme	Marks	AOs
	$\int (3x^{0.5} + A) dx = 2x^{1.5} + Ax(+c)$	M1 A1	3.1a 1.1b
	Uses limits and sets $= 2A^2 \Rightarrow (2 \times 8 + 4A) - (2 \times 1 + A) = 2A^2$	M1	1.1b
	Sets up quadratic and attempts to solve Sets up quadratic and attempts $b^2 - 4ac$	M1	1.1b
	$\Rightarrow A = -2, \frac{7}{2}$ and states that there are two roots States $b^2 - 4ac = 121 > 0$ and hence there are two roots	A1	2.4
(5 marks)			
Notes:			
M1: Integrates the given function and achieves an answer of the form $kx^{1.5} + Ax(+c)$ where k is a non-zero constant A1: Correct answer but may not be simplified M1: Substitutes in limits and subtracts. This can only be scored if $\int A dx = Ax$ and not $\frac{A^2}{2}$ M1: Sets up quadratic equation in A and either attempts to solve or attempts $b^2 - 4ac$ A1: Either $A = -2, \frac{7}{2}$ and states that there are two roots Or states $b^2 - 4ac = 121 > 0$ and hence there are two roots			

Q12.

Question	Scheme	Marks	AOs
(a)	$\int \frac{2}{(3x-k)} dx = \frac{2}{3} \ln(3x-k)$	M1	1.1a
		A1	1.1b
	$\int_k^{3k} \frac{2}{(3x-k)} dx = \frac{2}{3} \ln(9k-k) - \frac{2}{3} \ln(3k-k)$	dM1	1.1b
	$= \frac{2}{3} \ln\left(\frac{8k}{2k}\right) = \frac{2}{3} \ln 4$ oe	A1	2.1
		(4)	
(b)	$\int \frac{2}{(2x-k)^2} dx = -\frac{1}{(2x-k)}$	M1	1.1b
	$\int_k^{2k} \frac{2}{(2x-k)^2} dx = -\frac{1}{(4k-k)} + \frac{1}{(2k-k)}$	dM1	1.1b
	$= \frac{2}{3k} \left(\propto \frac{1}{k} \right)$	A1	2.1
		(3)	
(7 marks)			

(a) M1: $\int \frac{2}{(3x-k)} dx = A \ln(3x-k)$ Condone a missing bracket A1: $\int \frac{2}{(3x-k)} dx = \frac{2}{3} \ln(3x-k)$ Allow recovery from a missing bracket if in subsequent work $A \ln 9k-k \rightarrow A \ln 8k$ dM1: For substituting k and $3k$ into their $A \ln(3x-k)$ and subtracting either way around A1: Uses correct $\ln$ work and notation to show that $I = \frac{2}{3} \ln\left(\frac{8}{2}\right)$ or $\frac{2}{3} \ln 4$ oe (ie independent of k) (b) M1: $\int \frac{2}{(2x-k)^2} dx = \frac{C}{(2x-k)}$ dM1: For substituting k and $2k$ into their $\frac{C}{(2x-k)}$ and subtracting A1: Shows that it is inversely proportional to k Eg proceeds to the answer is of the form $\frac{A}{k}$ with $A = \frac{2}{3}$ There is no need to perform the whole calculation. Accept from $-\frac{1}{(3k)} + \frac{1}{(k)} = \left(-\frac{1}{3} + 1\right) \times \frac{1}{k} \propto \frac{1}{k}$ If the calculation is performed it must be correct. Do not isw here. They should know when they have an expression that is inversely proportional to k. You may see substitution used but the mark is scored for the same result. See below $u = 2x - k \rightarrow \left[\frac{C}{u} \right]$ for M1 with limits $3k$ and k used for dM1 For more help, please visit www.exampaperspractice.co.uk	
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